

```
In [5]: import numpy as np
```

```
marks=np.array([
    [85,78,90,88,76],
    [65,72,68,70,75],
    [92,88,95,90,94],
    [45,55,48,60,52],
    [78,82,75,80,85],
    [55,60,58,62,65],
    [88,91,84,87,90],
    [35,42,38,45,40],
    [70,68,72,75,71],
    [95,93,98,96,94]
])
```

```
In [3]: marks
```

```
Out[3]: array([[85, 78, 90, 88, 76],
               [65, 72, 68, 70, 75],
               [92, 88, 95, 90, 94],
               [45, 55, 48, 60, 52],
               [78, 82, 75, 80, 85],
               [55, 60, 58, 62, 65],
               [88, 91, 84, 87, 90],
               [35, 42, 38, 45, 40],
               [70, 68, 72, 75, 71],
               [95, 93, 98, 96, 94]])
```

```
In [3]: marks[:5]
```

```
Out[3]: array([[85, 78, 90, 88, 76],
               [65, 72, 68, 70, 75],
               [92, 88, 95, 90, 94],
               [45, 55, 48, 60, 52],
               [78, 82, 75, 80, 85]])
```

```
In [4]: marks[:,0]
```

```
Out[4]: array([85, 65, 92, 45, 78, 55, 88, 35, 70, 95])
```

```
In [5]: marks[0]
```

```
Out[5]: array([85, 78, 90, 88, 76])
```

```
In [6]: marks.sum(axis=1)
```

```
Out[6]: array([417, 350, 459, 260, 400, 300, 440, 200, 356, 476])
```

```
In [7]: marks.mean(axis=1)
```

```
Out[7]: array([83.4, 70. , 91.8, 52. , 80. , 60. , 88. , 40. , 71.2, 95.2])
```

```
In [8]: marks.max(axis=0)
```

```
Out[8]: array([95, 93, 98, 96, 94])
```

```
In [9]: marks.min(axis=0)
```

```
Out[9]: array([35, 42, 38, 45, 40])
```

```
In [10]: marks.max()
```

```
Out[10]: np.int64(98)
```

```
In [11]: marks.min()
```

```
Out[11]: np.int64(35)
```

```
In [12]: marks.mean(axis=1)[marks.mean(axis=1)>75]
```

```
Out[12]: array([83.4, 91.8, 80. , 88. , 95.2])
```

```
In [13]: marks[marks[:,0]>80,0]
```

```
Out[13]: array([85, 92, 88, 95])
```

```
In [14]: np.sum(marks.mean(axis=1)>=60)
```

```
Out[14]: np.int64(8)
```

```
In [16]: marks[marks<40]=40
```

```
In [17]: marks
```

```
Out[17]: array([[85, 78, 90, 88, 76],  
                [65, 72, 68, 70, 75],  
                [92, 88, 95, 90, 94],  
                [45, 55, 48, 60, 52],  
                [78, 82, 75, 80, 85],  
                [55, 60, 58, 62, 65],  
                [88, 91, 84, 87, 90],  
                [40, 42, 40, 45, 40],  
                [70, 68, 72, 75, 71],  
                [95, 93, 98, 96, 94]])
```

```
In [18]: marks_reshaped=marks.reshape(5,10)  
marks_reshaped
```

```
Out[18]: array([[85, 78, 90, 88, 76, 65, 72, 68, 70, 75],  
                [92, 88, 95, 90, 94, 45, 55, 48, 60, 52],  
                [78, 82, 75, 80, 85, 55, 60, 58, 62, 65],  
                [88, 91, 84, 87, 90, 40, 42, 40, 45, 40],  
                [70, 68, 72, 75, 71, 95, 93, 98, 96, 94]])
```

```
In [19]: marks_reshaped.shape
```

```
Out[19]: (5, 10)
```

```
In [20]: marks.shape[0]
```

```
Out[20]: 10
```

```
In [21]: marks.shape[1]
```

```
Out[21]: 5
```

```
In [22]: marks.shape
```

```
Out[22]: (10, 5)
```

<p>marks.size</p>

```
In [23]: marks.size
```

```
Out[23]: 50
```

```
In [5]: marks.flatten()
```

```
Out[5]: array([85, 78, 90, 88, 76, 65, 72, 68, 70, 75, 92, 88, 95, 90, 94, 45, 55,
              48, 60, 52, 78, 82, 75, 80, 85, 55, 60, 58, 62, 65, 88, 91, 84, 87,
              90, 35, 42, 38, 45, 40, 70, 68, 72, 75, 71, 95, 93, 98, 96, 94])
```

```
In [6]: marks.reshape(5,10)
```

```
Out[6]: array([[85, 78, 90, 88, 76, 65, 72, 68, 70, 75],
              [92, 88, 95, 90, 94, 45, 55, 48, 60, 52],
              [78, 82, 75, 80, 85, 55, 60, 58, 62, 65],
              [88, 91, 84, 87, 90, 35, 42, 38, 45, 40],
              [70, 68, 72, 75, 71, 95, 93, 98, 96, 94]])
```

```
In [7]: marks.reshape(5,10).shape
```

```
Out[7]: (5, 10)
```

```
In [6]: import numpy as np
```

```
mean=np.mean(marks)
median=np.median(marks)
std=np.std(marks)

print("Mean:",mean)
print("Median:",median)
print("Standard Deviation:",std)
```

Mean: 73.16

Median: 75.0

Standard Deviation: 17.505839025879336

```
In [9]: sorted_marks=np.sort(marks,axis=1)
print(sorted_marks)
```

```
[[76 78 85 88 90]
 [65 68 70 72 75]
 [88 90 92 94 95]
 [45 48 52 55 60]
 [75 78 80 82 85]
 [55 58 60 62 65]
 [84 87 88 90 91]
 [35 38 40 42 45]
 [68 70 71 72 75]
 [93 94 95 96 98]]
```

```
In [16]: total_marks=np.sum(marks,axis=1)

student=np.argmax(total_marks)

print("Total marks:",total_marks)
print("Highest total marks:",
      total_marks[student])
print("Student number:",student+1)
```

Total marks: [417 350 459 260 400 300 440 200 356 476]

Highest total marks: 476

Student number: 10

In []:

In []: